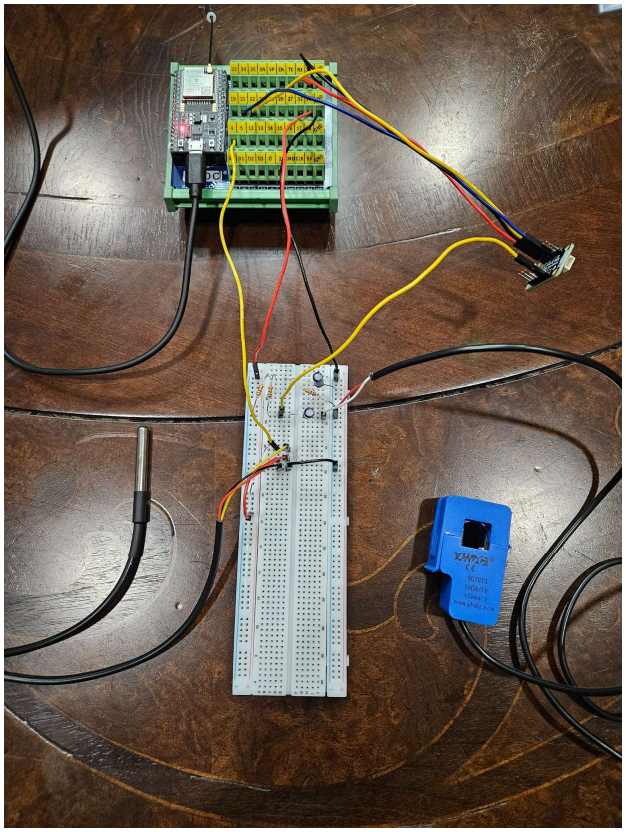
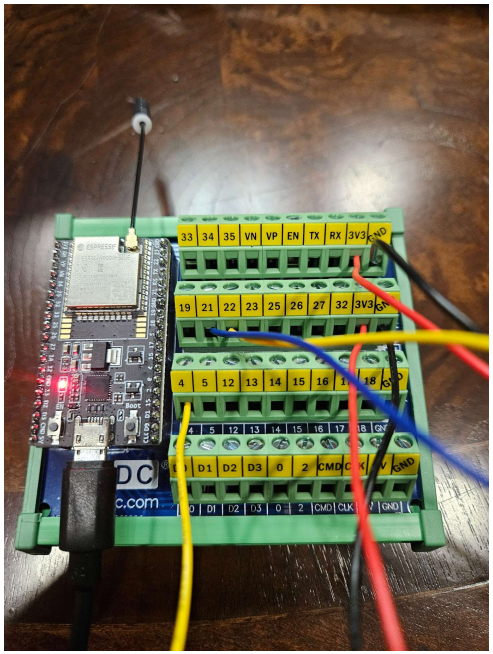
As-built prototype: photographic reference

Photographs provided October 31, 2026 • Pictures are referenced; captions included



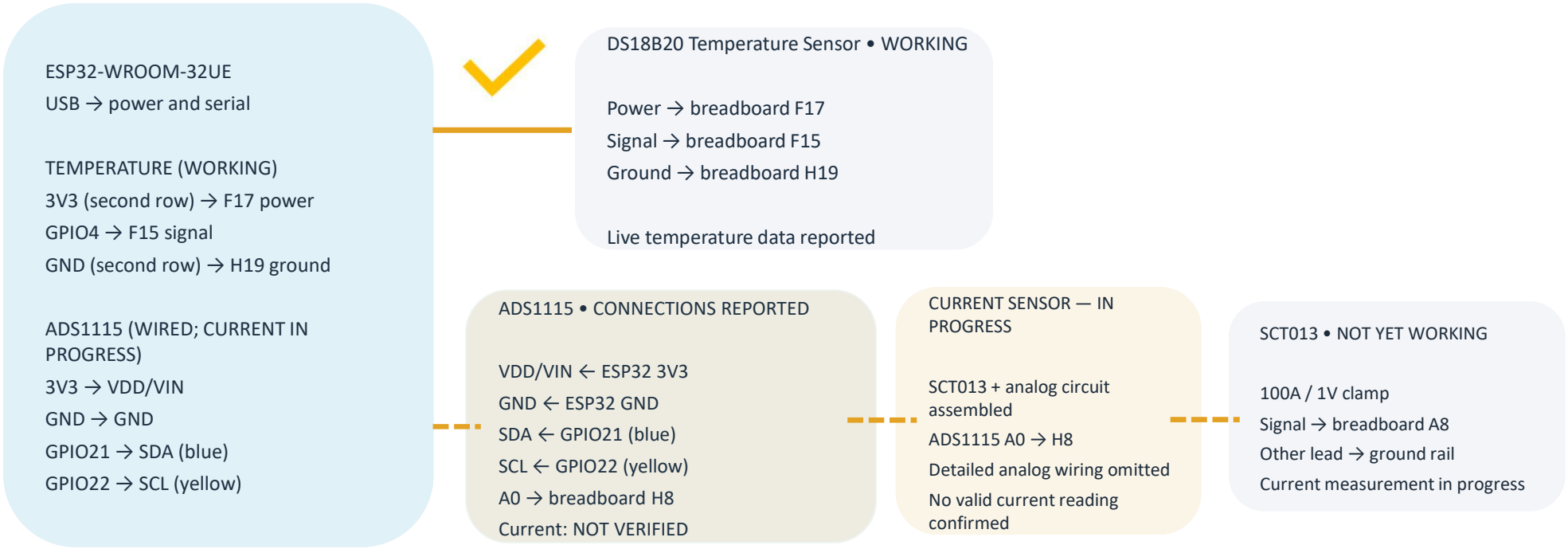


Visible hardware:

ESP32-WROOM-32UE • breadboard • DS18B20 Temperature Sensor • ADS1115 converter • SCT013 clamp (current measurement in progress)

Prototype wiring: Temperature monitoring

As-reported connections • Temperature: live readings • Current sensor: not yet working



System architecture & test status

SENSOR HARDWARE

DS18B20 Temperature Sensor — working

SCT013 + ADS1115 — in progress

CONTROLLER

ESP32 firmware

USB power and serial

DATA OUTPUT

Computer measurements

Website dashboard

EVIDENCE / TEST STATUS

✓ ESP32 powered and communicating with computer.

✓ DS18B20 Temperature Sensor connected to 3V3, GPIO4, and GND via breadboard.

✓ Live temperature readings reported in software and website.

✓ ADS1115 power and I²C wiring mapped (GPIO21 SDA; GPIO22 SCL).

□ Current-sensing circuit assembled, but no verified current readings yet.

□ Further testing: ADS analog input, breadboard wiring, and firmware.